

Aziz Ashfak

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RESEARCH INTERESTS

Machine Learning · Multimodal RAG · Retrieval-Augmented Generation · Causal Knowledge Extraction · LLM Evaluation · Low-Resource NLP

EDUCATION

B.Sc. (Honours) in Statistics, Noakhali Science and Technology University, Bangladesh *Apr 2022 – Aug 2026*
– **CGPA 3.37 / 4.00.**
– Thesis: BanglaStatMCQ-500, a statistical reasoning benchmark in Bangla. Supervisor: Md. Rasel Hossain, Assistant Professor, Department of Statistics.

RESEARCH EXPERIENCE

Research Collaborator (AI / ML), Gulf Council Research Project, University of South Alabama *Nov 2025 - Present*
– Project: *Evidence-grounded causal knowledge graphs from public fisheries discourse, with Mahi Washim.*
– Built a RAG pipeline over 3,000+ documents resolving 498,770 passages and 112,378 claims into a **fuzzy cognitive map of 909 evidence-weighted edges**.
– Audited it over twelve pre-registered experiments: **96.3% of claims traced to their cited source** against a 0.2% shuffled control, with the vocabulary's 86.1% discard rate and an 11.2-point judge-model swing reported alongside.

Research Collaborator, Coastal Evidence Synthesis Project, University of South Alabama *Jun 2026 - Present*
– Project: *Where coastal planning discourse diverges from the science it cites, with Mahi Washim.*
– Coded 8,940 segments from 103 PDFs (12,257 pages) at 94% audited precision and found **Public Health co-occurring below chance with every other theme** (lift 0.26 to 0.60).
– Recovered 744 cited references: **water quality fills 72.8% of documents but 2.9% of the science they cite.**

Research Data Collaborator, Human–Wildlife Conflict Corpus Project *Mar 2026 - Aug 2026*
– Project: *Comparative mental models of human–wildlife conflict across 19 South and Southeast Asian countries, with Mahi Washim.*
– Built a dual-pipeline ETL across 19 countries yielding 5,300 documents and **60,000+ provenance-linked concept nodes**, then compared ten partitioned maps by centrality, Mann-Whitney U, bootstrap and FDR correction.

PUBLICATIONS & MANUSCRIPTS

Accepted / Peer-Reviewed

[A1] “Optimizing U-Net Performance: Evaluation of State-of-the-Art Encoders and Optimizers on Human Segmentation Dataset.” *IEEE SMC 2026, Bellevue, WA. [co-author; to appear in IEEE Xplore]*

Under Review

- [R1] “Robustness of CNN Transfer Learning versus Vision Transformers for Aerial Scene Classification.” *ICCIT 2026. [first author]*
- [R2] “Self-Supervised versus ImageNet Pretraining for Bangladeshi Traffic Vehicle Recognition under Limited Labels.” *IEEE BDCloud 2026. [first author]*
- [R3] “Transfer Learning for Chest X-Ray Lung Segmentation: A Comparative Study of CNN Backbones.” *Scientific Reports. [co-author]*
- [R4] “Skilled Birth Attendance and Height-for-Age Among Children Aged 0–23 Months in Bangladesh: A Survey-Weighted TMLE Analysis of the 2025 MICS.” *BMC Public Health. [co-author]*
- [R5] “Functional Difficulty and ICT Skill Attainment Among Women in Bangladesh: Evidence from the 2025 MICS.” *BMC Public Health. [co-author]*
- [R6] “Same Score, Different Deprivation: A Survey-Weighted Latent Class Analysis of Multidimensional Deprivation Among Children Aged 24–59 Months.” *BMC Public Health. [co-author]*

In Preparation

[P1] “Can a Retrieval-Augmented Fuzzy Cognitive Map Be Trusted? A Twelve-Experiment Audit of Gulf of Mexico Fisheries Discourse.” *[co-author]*

[P2] “Retrieval Grounds Causal Maps: A Benchmark for Fuzzy Cognitive Maps Extracted from Language-Model Answers.” *[co-author]*

[P3] “Comparative Mental Models of Human–Wildlife Conflict across South and Southeast Asia: A Fuzzy Cognitive Map Analysis.” *[co-author]*

[P4] “AI-Mined Fuzzy Cognitive Maps for Evidence-Grounded Wildlife Conservation in the U.S. Southwest.” *[co-author]*

[P5] “Evaluating Multimodal StatTransformer Architectures with Large Language Models for Statistical Reasoning.” *[first author]*

PROFESSIONAL EXPERIENCE

AI Engineer, ROYALX LLC, Dhaka, Bangladesh (remote, full-time) *Mar 2025 - Nov 2025*

- Shipped **LLM applications and RAG pipelines** (OpenAI, Groq, Hugging Face, LangChain, LlamaIndex) as production FastAPI and Streamlit services, and fine-tuned transformers for question answering and document understanding.

AI Tutor, AI Devlop, Dhaka, Bangladesh (remote, part-time) *Apr 2025 - Aug 2025*

- Taught **100+ students** exploratory analysis, statistics, machine learning and deep learning through end-to-end ML pipelines.

Freelance Data Analyst, Hestia Traffic App Project *Dec 2024 - Feb 2025*

- Modelled **crash severity** against violation, exposure and environmental variables on large-scale road-accident data.

SELECTED PROJECTS

Multimodal Research Paper Analyst, multimodal RAG over paper text and figures *[code] [demo]*

- Groq and Llama pipeline that **summarises text and interprets charts**; Streamlit app exporting to Word, PDF, CSV and Markdown.

Brain Tumor Detection, **98.52%** accurate MRI tumour classifier *[code] [live]*

- LeNet, ResNet152V2 and DenseNet121 compared under matched augmentation, winner served as an ONNX service with CI/CD.

Indian Flight Price Prediction, **99.69%** accurate fare model over 300,261 records *[code] [live]*

- BaggingRegressor behind a reproducible preprocessing pipeline, deployed through GitHub Actions.

QnA RAG Bot, context-grounded question answering over uploaded PDFs *[code]*

- FastAPI, LangChain and Hugging Face with **FAISS vector search** and figure analysis.

TECHNICAL SKILLS

Programming	Python , SQL, R, C (basic), Java (basic)
ML / DL	PyTorch , TensorFlow, scikit-learn, Transformers, Vision Transformers, CNNs, transfer learning, PEFT/LoRA
LLMs & Retrieval	LangChain , LangGraph, LlamaIndex, OpenAI API, Hugging Face, FAISS, ChromaDB, multi-modal and agentic RAG
Causal & Graph	Fuzzy cognitive maps , causal knowledge-graph extraction, provenance tracking, centrality and co-occurrence analysis
Statistics	Hypothesis testing , regression, experimental design, bootstrap resampling, permutation tests, FDR correction, benchmark design
Engineering	PostgreSQL , Docker, FastAPI, Flask, ONNX, Git and GitHub Actions CI/CD, ETL pipeline design

REFERENCES

Mahi Washim Graduate Research Assistant University of South Alabama, Mobile, AL, USA mw2534@jagmail.southalabama.edu	Md. Rasel Hossain Assistant Professor, Department of Statistics Noakhali Science and Technology University, Bangladesh rasel.stat@nstu.edu.bd
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